MIS272 – Predictive Analytics

T2 2023

Assignment 2 – Individual
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Executive summary (1 page)

AirBnB encounters a business problem that revolves around analysing and improving the customer experience for AirBnB's domestic tourism offerings in Melbourne. This includes understanding the impact of property location on reviews, estimating review scores using predictive models, and identifying the scores of properties with review scores that are concerning due to its inability to meet specific standards. The solutions produced through predictive modelling will assist AirBnB to make data-driven decision which will ultimately strengthen its services and customer satisfaction.

The primary goal is to elevate AirBnB's operations in the domestic tourism sector by improving property quality, customer satisfaction and the overall booking experience. Achieving such objectives will benefit the property hosts as well as the guests which can be achieved with data analysis and predictive modelling.

In Task A, AirBnB are seeking insights into prices or properties that are more than ten kilometres from the Melbourne CBD. Shown in Figure 1.5, there is not a meaningful relationship between each property's review score rating and the distance from the Melbourne CBD which is explained by the lack of correlation of 0.048**.** The following task estimates the review score rating for all properties located more than ten kilometres from the Melbourne CBD. Predictive modelling (see Figure 2.1) was used to help predict results of what factors lead to a higher review score rating. In the final task, we identify review scores that do not meet a certain criterion and specify the number of properties in a segment. After developing a process model (see Figure 6.1), conclusions were made that the second segment ('cluster_1', as shown in Figure 3.4) was the most concerning as it resulted in the greatest average within the centroid distance of 0.356 and the segment with the greatest number of properties was the third segment ('cluster_2', as shown in Figure 3.3)

After the development of a process model, suggestions can be made to AirBnB which include to consider the impact of property location (distance from the Melbourne CBD) on review scores when hosts set prices for their AirBnB property. Furthermore, employing the predictive model that will help estimate review score ratings is a beneficial asset to AirBnB which can allows hosts to assess the strengths and weaknesses of their property. Ultimately, these improvements can enhance the guest's experience. A final recommendation to include is that although AirBnB have been notified with properties with review scores below 10, it is pivotal to not shift focus on the properties with review scores of 10 as we can gain insight from the strengths of these properties.

By implementing data-driven insights and predictive models, AirBnB can improve the quality of their listings and maintain a high standard of customer satisfaction.

Data exploration, pattern discovery, and preparation (2 pages)

TASK A

In task A, we were asked to determine whether a correlation exists between two attributes being the property's calculated distance to Melbourne's CBD and each property's review score rating. To interpret these results, a process model must be created which requires modifications to the dataset in order to acquire an accurate reading. As shown in figure 1.1, selected attributes are latitude, longitude and review_scores_rating. The latitude and longitude attributes are selected to generate a new attribute that is named Distance_to_CBD. Displayed in figure 1.3 below, is the formula that is applied and works out the distance to the Melbourne CBD from each property. The chosen set role attribute name, as shown in Figure 1.4, is review_scores_rating undertaking a target role of "Label" which is used to specify the attribute that you want to predict or model. After conducting regression analysis, the coefficient of determination between the two attributes came to a result of 0.048. This R-squared value represents a weak linear relationship and a weak effect on explaining the changes between these two variables.

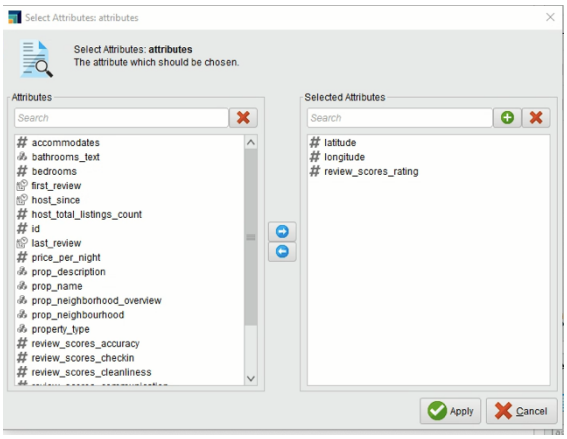


Figure 1.1

Figure 1.2

Figure 1.3

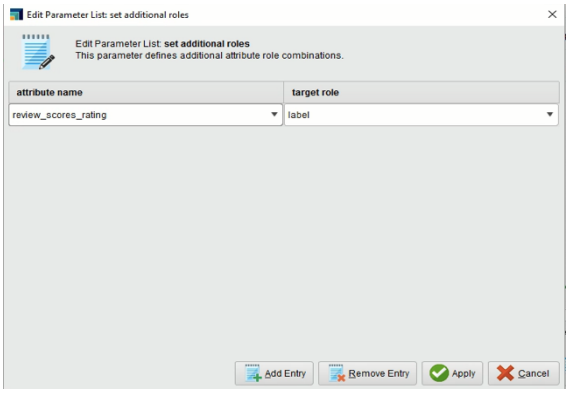


Figure 1.4

squared_correlation

```
squared_correlation: 0.048 +/- 0.032 (micro average: 0.038)
```

Figure 1.5

Task B

Task B's objective was to develop a predictive model to estimate the review score ratings. After the dataset was imported, any missing value was filtered and removed to improve the reliability of the results. Throughout the process model, attributes were picked that would be the relevant predictors of the review score ratings (shown in Figure 2.1). The output for the regression formula is shown in figure 2.2. Observing the linear regression table in figure 2.3, we can see that the p-Value for the selected attributes is less than 0.05. This implies that the coefficient for that particular predictor variable is statistically significant. The correlation matrix, listed as figure 2.4, gives us an insight to whether multicollinearity is present in our model. We recognise a correlation between 'review_scores_cleanliness' and 'review_scores_rating' which could affect the interpretability of the regression model.

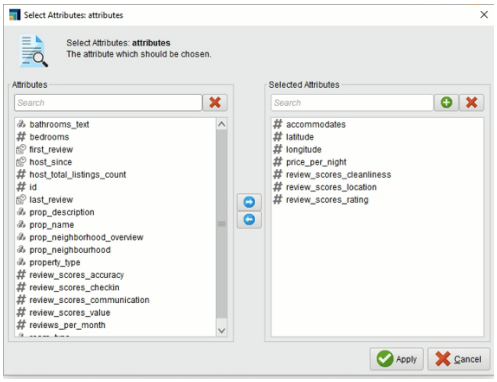


Figure 2.1

LinearRegression

```
- 0.300 * accommodates
+ 0.006 * price_per_night
+ 4.359 * review_scores_cleanliness
+ 3.506 * review_scores_location
+ 19.902
```

Figure 2.2

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
accommodates	-0.300	0.073	-0.184	0.956	-4.105	0.000	****
price_per_night	0.006	0.002	0.113	0.985	2.588	0.010	**
review_scores_cleanliness	4.359	0.306	0.597	0.888	14.253	0	****
review_scores_location	3.506	0.574	0.253	0.903	6.105	0.000	****
(Intercept)	19.902	5.698	?	?	3.493	0.001	****

Figure 2.3

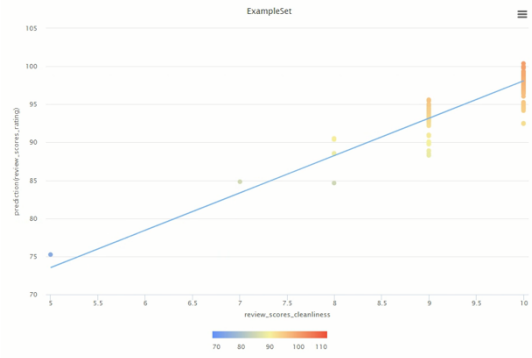


Figure 2.5

Attributes	latitude	longitude	accommodates	price_per_night	review_scores_rating	review_scores_cleanliness	review_scores_location ↑
accommodates	-0.021	-0.025	1	0.489	-0.189	-0.220	-0.074
price_per_night	0.009	0.301	0.489	1	0.063	0.025	-0.008
latitude	1	0.157	-0.021	0.009	0.030	0.047	0.019
longitude	0.157	1	-0.025	0.301	0.192	0.195	0.058
review_scores_cl...	0.047	0.195	-0.220	0.025	0.716	1	0.286
review_scores_rat...	0.030	0.192	-0.189	0.063	1	0.716	0.462
review_scores_lo...	0.019	0.058	-0.074	-0.008	0.462	0.286	1

Figure 2.4

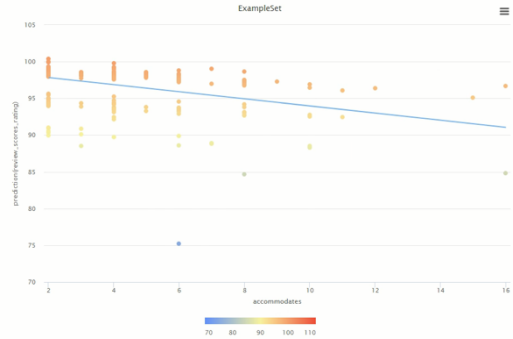


Figure 2.6

TASK C

Task C asks to develop a process model to identify properties of concern based on the six review score attributes. Figure 3.1 illustrates the list of review score attributes chosen. A filter was used on these attributes to ensure that review scores only less than 10 were used for this process model which is explained in figure 3.2. The results from the model produced a cluster model which interprets a total number of 22 items (figure 3.3). The Davies-Bouldin index value is at 0.729 which is at an interpretable figure meaning that the clusters are well separated. Visualised in the performance vector in figure 3.4, it is clear that the average within the centroid distance in ‘cluster_1’ has the greatest value at 0.356 which results in this cluster being the most concerning.

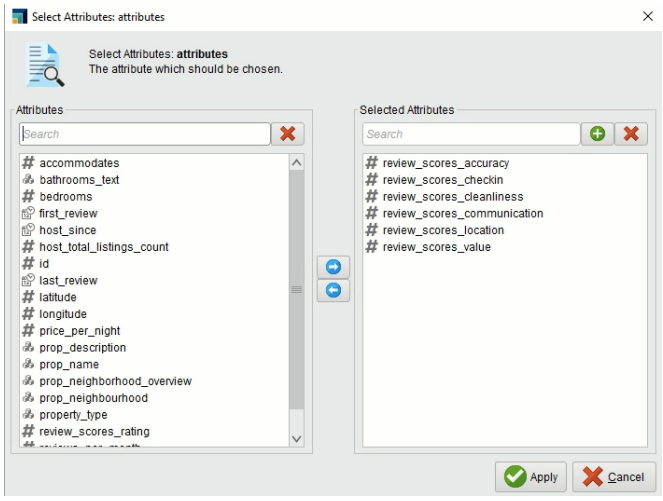


Figure 3.1

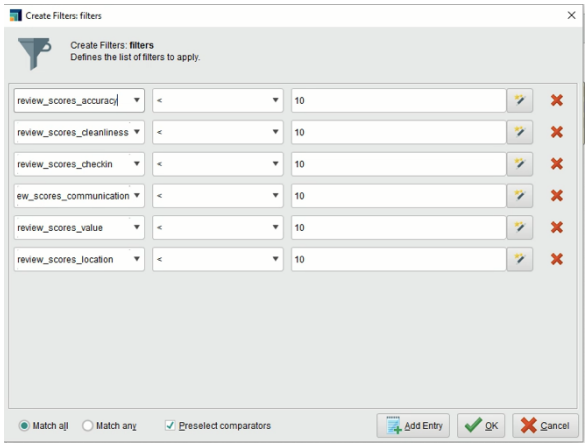


Figure 3.2

Cluster Model

```
Cluster 0: 5 items
Cluster 1: 2 items
Cluster 2: 15 items
Total number of items: 22
```

Figure 3.3

PerformanceVector

```
PerformanceVector:
Avg. within centroid distance: 0.120
Avg. within centroid distance_cluster_0: 0.139
Avg. within centroid distance_cluster_1: 0.356
Avg. within centroid distance_cluster_2: 0.082
Davies Bouldin: 0.729
```

Figure 3.4

Predictive modelling (2 pages)

In Task A, we selected the attributes latitude and longitude so that for the future step we are able to use these attributes to calculate the 'Distance_to_CBD' using geolocational data. The final operator, being cross validation (figure 4.1), is used to assess the performance of the predictive model. Figure 4.3 shows the parameters for the linear regression operator. The t-test selection must be chosen so that the model is able to distinguish whether the coefficients are statistically significant.

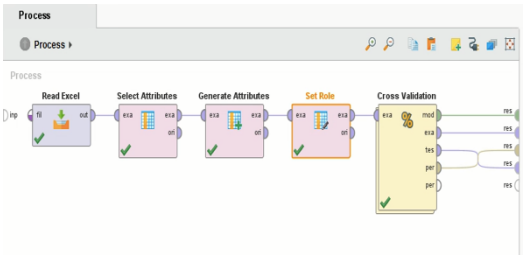


Figure 5.1

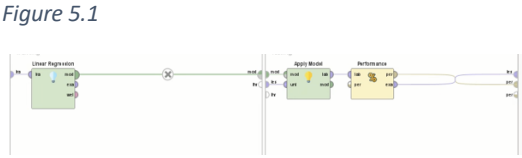


Figure 4.2

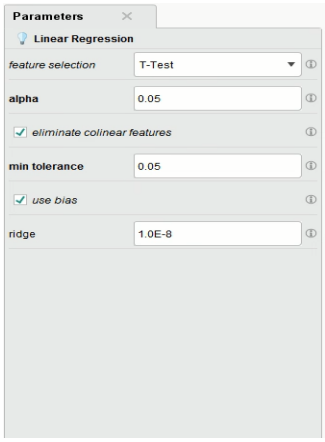


Figure 4.3

Selected attributes such as 'review_scores_cleanliness' and 'accommodates' were chosen as it was believed that the higher that these attributes were within their scale, the greater the review score rating would be. However, as shown in figure 2.6, the regression analysis shows otherwise. The correlation matrix operator is solely there to identify multicollinearity within the regression analysis which ultimately ensures the validity within the predictive model.

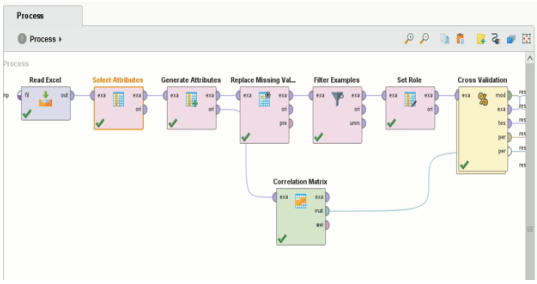


Figure 5.1

In relation to the question of task c, the focus is on the review scores that are below 10 for listed properties across the dataset. To satisfy this, we appoint the 'filter examples' operator and create entries for each attribute and set them to 'less than (<) 10'. The next step in this process model is to apply the 'normalize' operator (figure 6.1). Using this enables us to compare the data that were originally in different scales. If we look at figure 6.2, we can address that the minimum values for certain attributes have different values requiring this operator. The following operator added is 'clustering' which requires a k value of 3 in the parameters (figure 6.3).

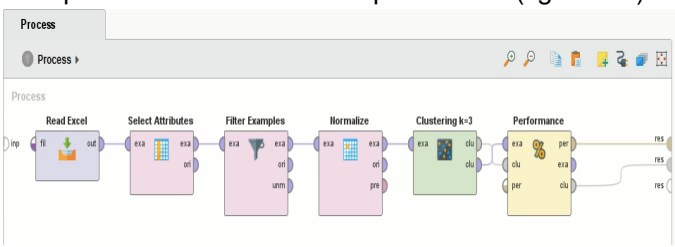
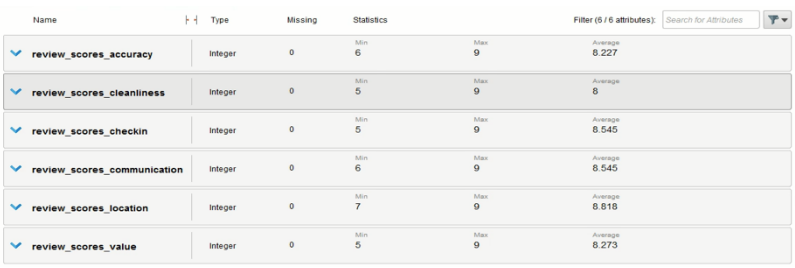


Figure 6.1



Name	Type	Missing	Statistics		
			Min	Max	Average
review_scores_accuracy	Integer	0	6	9	8.227
review_scores_cleanliness	Integer	0	5	9	8
review_scores_checkin	Integer	0	5	9	8.545
review_scores_communication	Integer	0	6	9	8.545
review_scores_location	Integer	0	7	9	8.818
review_scores_value	Integer	0	5	9	8.273

Figure 6.2

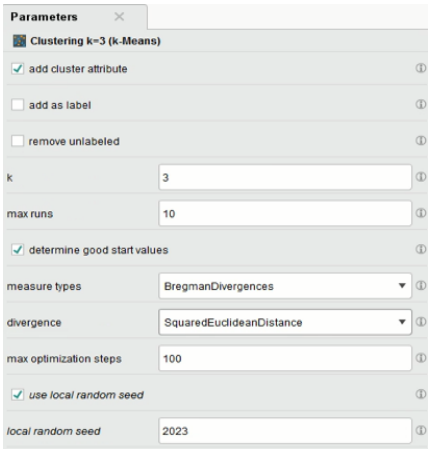

Parameters
Clustering k=3 (k-Means)
add cluster attribute
add as label
remove unlabeled
k
3
max runs
10
determine good start values
measure types
BregmanDivergences
divergence
SquaredEuclideanDistance
max optimization steps
100
use local random seed
local random seed
2023

Figure 6.3

Model evaluation and improvement (2 pages)

The cross validation operator was employed to enhance the accuracy of the results in task A and task B. In task B, we adopted the visualisation of a scatterplot (figure 2.5 and 2.6) to appropriately show linear regression and assess the relationship between two variables. Furthermore, the correlation matrix operator was used which provided an insight into whether there were any selected attributes that were inappropriate to use. Shown in figure 7.1, we can see on the matrix that the variables 'accommodates' and 'bedrooms' have a strong relationship based on their figure of 0.847. This implies multicollinearity and therefore requires attention. To confirm this, we can see in the linear regression table that the p-Value for the two attributes is greater than 0.05 (figure 7.2). This tells us that the predictor variable is not statistically significant in explaining the variation in the dependant variable. Removing the attribute, 'bedrooms', allowed the model to work without the issue of multicollinearity which was further confirmed by consistent p-Value's under 0.05.

Attributes	latitude	longitude	accommodates	bedrooms	price_per_night	review_scores_rating	review_scores_cleanliness	review_scores_location
latitude	1	0.157	-0.021	-0.018	0.009	0.030	0.047	0.019
longitude	0.157	1	-0.025	0.053	0.301	0.192	0.195	0.058
accommodates	-0.021	-0.025	1	0.847	0.489	-0.189	-0.220	-0.074
bedrooms	-0.018	0.053	0.847	1	0.538	-0.139	-0.128	-0.087
price_per_night	0.009	0.301	0.489	0.538	1	0.063	0.025	-0.008
review_scores_rating	0.030	0.192	-0.189	-0.139	0.063	1	0.716	0.462
review_scores_cleanliness	0.047	0.195	-0.220	-0.128	0.025	0.716	1	0.286
review_scores_location	0.019	0.058	-0.074	-0.087	-0.008	0.462	0.286	1

Figure 7.1

Attribute	Coefficient	Std. Error	Std. Coefficient	Tolerance	t-Stat	p-Value	Code
accommodates	-0.180	0.142	-0.110	0.870	-1.270	0.205	
bedrooms	-0.348	0.354	-0.086	0.856	-0.982	0.327	
price_per_night	0.006	0.002	0.118	0.984	2.686	0.008	***
review_scores_cleanli...	4.342	0.306	0.595	0.885	14.174	0	****
review_scores_location	3.463	0.576	0.250	0.900	6.012	0.000	****
(Intercept)	20.603	5.743	?	?	3.587	0.000	****

Figure 7.2

Ultimately, AirBnB is able to use predictive modelling to enhance customer satisfaction based on these results. We can see an example of this in figure 2.5 explaining that a cleaner property is explained by the increase in a higher customer score review. These data-driven decisions are pivotal to make based off the accurate models generated.